Visual Analytics for Global Refugee Migration Patterns Using Python

https://www.kaggle.com/datasets/unitednations/refugee-data

Student Name:

Student ID:

Supervisor Name:

ACKNOWLEDGEMENT

The successful completion of this dissertation was made possible through the guidance, resources, and academic support provided by the faculty and supervisors. Gratitude is extended to the University for providing access to research facilities and datasets. Appreciation is also expressed to family and peers for their encouragement throughout the research journey.

DECLARATION

This dissertation is a unique work and it has not been registered in any academic institution before. The references and all sources of the information have been mentioned properly. The study did not violate the rules of academic integrity or ethical research conduct.

ABSTRACT

In this research, a Python-based visual analytics and predictive forecasting model was created to examine the trends of refugee migration in the world. The study was aimed at discovering the temporal and spatial dynamics, predicting the future trends of refugees and providing visibility insights in the form of interactive visual dashboard to aid humanitarian planning and policy-making. Data on migrants and refugees obtained by the United Nations and Kaggle repositories were imported, cleaned and pre-processed with the help of Python libraries: pandas and NumPy. Matplotlib, Seaborn, and Plotly were used to perform the exploratory data analysis to reveal the historical trends. ARIMA, Prophet and LSTM forecasting models were applied to make future predictions of refugee movement. Visualization and predictive output were incorporated by creating an interactive dashboard to integrate visualization and predictive outputs.

The analysis showed that there are long-term growth patterns and high temporal changes in the population of refugees in the world. The LSTM model had the best predictive accuracy and then ARIMA and Prophet had a relatively low predictive accuracy. Visual analytics increased interpretation and understanding. The research results in a conclusion that predictive analytics and interactive dashboards written in Python offer useful means of comprehending the patterns of refugee migration and facilitating the process of making humanitarian decisions based on evidence.

Table of Contents

ACKNOWLEDGEMENT 2

DECLARATION 3

ABSTRACT 4

Chapter 1: Introduction 10

1.1 Background of the Study 10

1.2 Research Rationale 10

1.4 Problem Statement 11

1.5 Significance of the Study 11

1.6 Structure of the Dissertation 12

Chapter 2: Literature Review 13

2.1 Introduction 13

2.2 Literature on Refugee Data and Visual Analytics 13

2.2.1 Availability and Evolution of Refugee Datasets 13

2.2.2 Use of Visual Analytics in Humanitarian Contexts 14

2.2.3 Limitations in Current Refugee Visualisation Systems 15

2.3 Predictive Modelling in Refugee Migration 16

2.3.1 Time-Series Forecasting Models 16

2.3.2 Machine Learning Approaches 17

2.3.3 Integration of Visualization and Prediction 18

2.4 Data Challenges in Refugee Analytics 19

2.4.1 Data Quality Issues 19

2.4.2 Data Integration and Standardisation 20

2.4.3 Ethical and Privacy Considerations 20

2.5 Gaps in Existing Literature 21

2.6 Conclusion 22

Chapter 3: Research Aim and Objectives 23

3.1 Research Aim 23
3.2 Research Objectives 23
3.3 Research Questions 23
Chapter 4: Methods and Methodology 24
4.1 Introduction 24
4.2 Research Design 24
4.3 Data Collection 25
4.4 Data Preprocessing 25
4.5 Analytical Methods 26
4.6 Model Development 26
4.7 Dashboard Development Using Streamlit 27
4.8 Ethical Considerations 27
4.9 Conclusion 27
Chapter 5: Results Analysis, Implementation, and Development 29
5.1 Result Analysis 29
5.1.1 Dataset Overview and Contextual Understanding 29
5.1.2 Data Loading and Initial Inspection 32
5.1.3 Data Cleaning and Missing Value Handling 33
5.1.4 Temporal Aggregation and Global Trend Formation 34
5.1.5 Exploratory Data Analysis and Visual Interpretation 35
5.1.6 Feature Engineering and Time-Series Structuring 39
5.2 Implementation and Development 39
5.2.1 System Environment and Library Configuration 39
5.2.2 Data Pipeline and Preprocessing Workflow 41
5.2.3 ARIMA Model Implementation 42
5.2.4 Prophet Model Implementation 43
5.2.5 LSTM Model Development 45

5.2.6 Model Integration and Comparative Framework 46
5.2.7 Interactive Dashboard Development 47
5.3 Conclusion 48
Chapter 6: Evaluation 49
6.1 Introduction 49
6.2 Evaluation Framework and Experimental Setup 49
6.3 Evaluation Metrics and Performance Criteria 50
6.4 ARIMA Model Evaluation 50
6.5 Prophet Model Evaluation 51
6.6 LSTM Model Evaluation 52
6.7 Comparative Model Performance Analysis 53
6.8 Dashboard Evaluation and Decision-Support Effectiveness 54
6.9 Discussion 55
6.9.1 Model Accuracy and Reliability 55
6.9.2 Trade-offs Between Interpretability and Performance 55
6.9.3 Practical Implications for Humanitarian Planning 55
6.10 Conclusion 56
Chapter 7: CONCLUSION 57
7.1 Main Restrictions and Problems 57
7.2 Achievement of Aim and Objectives 58
7.3 Achievement of Research Questions 58
7.4 Key Contributions 59
7.5 Lessons Learned (Reflection) 60
7.7 Final Remark 61
References 62

List of Figures

Figure 1Twitter activity of likely refugees moving from the Middle East to Germany 16
Figure 2 LSTM(Long Short-Term Memory) Approach 18
Figure 4 Methodology Flowchart: Visual Analytics for Refugee Migration..... 30
Figure 5 Extracting dataset from Kaggle 31
Figure 6 Dataset Overview 31
Figure 7Data Loading and Initial Inspection 32
Figure 8 Checking missing values 33
Figure 9 Filtering refugee population 34
Figure 10EDA Results..... 35
Figure 11EDA Results..... 36
Figure 12EDA Results..... 36
Figure 13EDA Results..... 37
Figure 14EDA Results..... 37
Figure 15EDA Results..... 38
Figure 16EDA Results..... 38
Figure 17System Environment and Library Configuration..... 40
Figure 18 Data preprocessing 41
Figure 19 Data preprocessing 42
Figure 20 Data preprocessing 42
Figure 21ARIMA Model Implementation..... 43
Figure 22 PROMPHET Model Implementation 44
Figure 23 LSTM Model Implementation 45
Figure 24 Model Comaprison Implementation..... 46
Figure 25 Dashboard Implementation..... 47
Figure 26 Results: ARIMA Model 51
Figure 27 Results: PROPHET Model 52
Figure 28 Results: LSTM Model 53
Figure 29 Model Comparison..... 53
Figure 30 Refugee Migration Decision Support Dashboard 54

List of tables:

Table 1Structure of the Dissertation 12
Table 2Evolution of Refugee Datasets 14
Table 3Use of Visual Analytics in Humanitarian Contexts 15
Table 4Predictive Modelling Approaches in Refugee Migration 16
Table 5 LSTM Architecture Parameter Definitions 18
Table 6Common Data Quality Issues in Refugee Datasets 20
Table 7Ethical and Privacy Requirements in Refugee Analytics 21
Table 8Identified Gaps in Existing Refugee Analytics Literature 22

Chapter 1: Introduction

1.1 Background of the Study

The global migration of refugees has emerged to be a critical fact in modern humanitarian and socio-economic realities as a result of wars, persecution, political instability, and climate-related catastrophes. Each year, millions of people are forced out, causing multifaceted international, social, and economic complications. Proper and prompt knowledge of these migration trends is important in ensuring that the governments, international bodies, and non-governmental organizations (NGO) are coordinated to respond effectively in humanitarian activities. Although data on refugees at a considerable scale can be processed by other organizations like the United Nations High Commissioner of Refugees (UNHCR) and the International Organization of Migration (IOM), among others, the conventional methods of analysis usually paint a picture of a static view and do not effectively depict the dynamics within time and space. Such limitations negatively affect the possibility to foresee new migration hotspots and recognize areas that are at increased risk of displacement.

The recent developments of data analytics and visualization provide a chance to solve these issues. Python is a flexible and open-source programming platform that offers comprehensive libraries to work with data and manipulate it as well as visualize and even learn how to make actionable information based on the complex datasets. Pandas, NumPy, Matplotlib, Seaborn, and Plotly are libraries that help researchers to clean, preprocess, analyze, and visualize tremendous amounts of refugee data. Some of the time-series predictive models that make it possible to foresee the future trajectory of migration flows include ARIMA, Prophet, and Long Short-Term Memory (LSTM) networks, which increase the chances of anticipating future changes. It is believed that by combining these tools, the study will develop a holistic analysis framework that is not only able to visualize the current migration trends but can also forecast the future displacement pattern to impact future policy and humanitarian efforts.

1.2 Research Rationale

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

Chapter	Title	Description
Chapter 1	Introduction	Presents study background, rationale, problem statement, significance, aim, objectives, and research questions guiding the investigation.
Chapter 2	Literature Review	Reviews existing studies on hospital readmissions, healthcare analytics, machine learning models, data challenges, and predictive methodologies.
Chapter 3	Requirements Specification	Identifies dataset requirements, preprocessing needs, modelling constraints, and key expectations for developing the predictive ML framework.
Chapter 4	Analysis and Design	Explains dataset characteristics, feature engineering, model selection, analytical workflow, and system design for predictive modelling.
Chapter 5	Implementation and Testing	Describes preprocessing procedures, model training, evaluation metrics, optimization steps, and comparative testing of regression and classification models.
Chapter 6	Demonstration and Evaluation	Presents prediction results, visual outputs, interpretability findings, model comparisons, and analytical performance evaluations.
Chapter 7	Conclusion	Summarizes findings, discusses implications, highlights study limitations, and suggests areas for future research in predictive healthcare analytics.

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

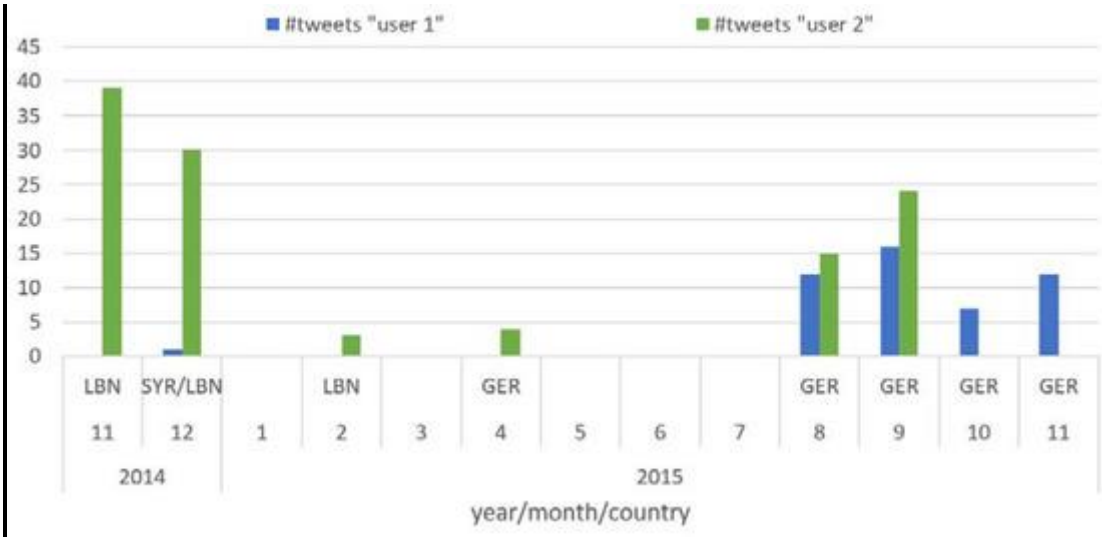
Source/Organisation	Type of Data Provided	Strengths	Limitations
UNHCR	Displacement, asylum, demographics	Global coverage, long historical span	Variations in reporting formats
IOM	Migration flows, border movements	Field-based data collection	Inconsistent country labelling
World Bank	Socioeconomic and population data	Supports cross-sector analysis	Time gaps in certain regions
Open-Data Initiatives	Crowdsourced and curated datasets	High transparency, open access	Quality varies across contributors
Regional Agencies	Local displacement records	Micro-level insights	Lack of standardisation

[redacted]

[redacted]

Visual Method	Main Purpose	Advantages	Limitations
Dashboards	Operational monitoring	Quick situational awareness	Often static and non-interactive
Heat Maps	Spatial distribution	Easy interpretation of hotspots	Limited contextual detail
Flow Maps	Migration path visualisation	Shows direction & volume of movement	Requires high-quality geodata
Time-Series Charts	Trend analysis	Effective for long-term tracking	Not ideal for multidimensional data
Interactive Interfaces	Dynamic exploration	Supports real-time insights	Requires technical skill to build

[redacted]



[redacted]

[redacted]

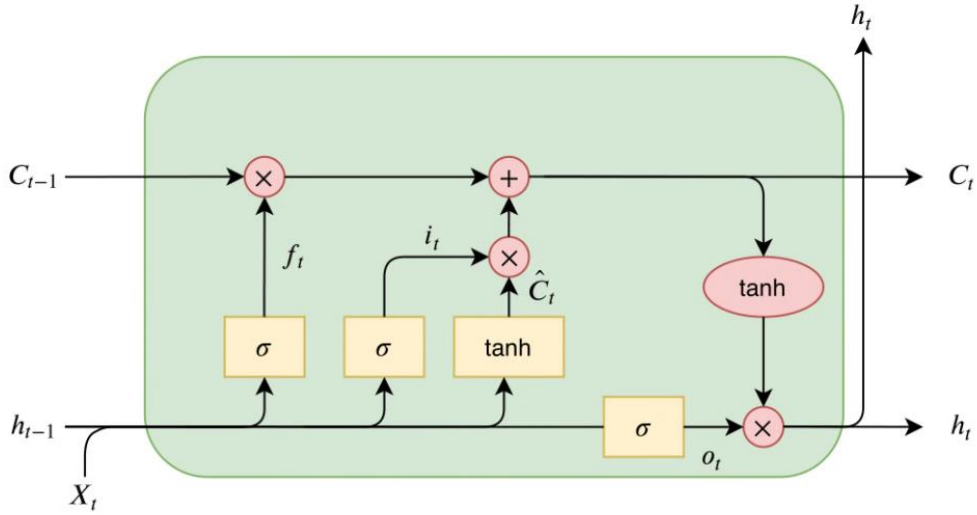
Model Type	Key Features	Strengths	Weaknesses
ARIMA	Trend and seasonal forecasting	Good for structured time patterns	Struggles with non-linear variations
Prophet	Automated trend detection	User-friendly and flexible	Depends heavily on data quality
LSTM	Long-term dependency detection	Captures complex temporal dynamics	Requires large datasets
Clustering Models	Group pattern identification	Useful for segmentation	Sensitive to inconsistent data
Hybrid Models	Combines ML + statistical methods	Enhances forecasting accuracy	Complex to implement

[redacted]

[redacted]

[redacted]

[redacted]



[redacted]

[redacted]

Parameter	Name	Description
X_t	**Input Vector**	The refugee migration data or features for the current time step t.
C_{t-1}	**Previous Cell State**	The "long-term memory" carrying information from all previous steps.
h_{t-1}	**Previous Hidden State**	The "short-term memory" or the output of the previous LSTM cell.
f_t	**Forget Gate**	Decides which information from C_{t-1} should be discarded.
i_t	**Input Gate**	Determines which new information will be stored in the cell state.
$\hat{C}_t$	**Candidate Cell State**	A vector of new candidate values to be added to the cell state.
C_t	**Current Cell State**	The updated long-term memory after applying the gates.
o_t	**Output Gate**	Decides which parts of the cell state will be output to the hidden state.
h_t	**Current Hidden State**	The final output at time t, used for the next step and the forecast.

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

Data Issue	Description	Impact on Analysis	Required Preprocessing
Missing Entries	Absent demographic or location fields	Reduces model reliability	Imputation or exclusion
Inconsistent Labels	Different country or category naming	Causes merging problems	Standardisation workflows
Duplicate Records	Repeated case entries	Inflates trends artificially	Deduplication
Time Gaps	Missing reporting periods	Breaks temporal continuity	Interpolation or smoothing
Incomplete Metadata	Lack of source descriptions	Reduces trust and traceability	Manual verification

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

Ethical Area	Requirement	Purpose	Risks if Neglected
Anonymisation	Remove personal identifiers	Protect individual identity	Exposure of vulnerable groups
Data Aggregation	Group-level reporting	Prevent re-identification	Misuse of sensitive patterns
Consent & Transparency	Clear data usage guidelines	Improve accountability	Loss of public trust
Secure Storage	Protected access systems	Prevent data breaches	Harm to displaced populations
Ethical Publication	Responsible dissemination	Avoid stigmatisation	Misinterpretation by stakeholders

[redacted]

[redacted]

[redacted]

Gap Area	Description	Impact on Research	Opportunity for Improvement
Lack of Integration	Visualization and prediction used separately	Limited analytical depth	Develop combined platforms
Proprietary Systems	Closed dashboards dominate	Low adaptability	Promote open-source tools
Low Interactivity	Static charts widely used	Limited user exploration	Build dynamic interfaces
Minimal Demographic Inputs	Models ignore age, gender, sectoral factors	Poor real-world accuracy	Add multidimensional variables
Limited Standardisation	Diverse data formats	Hinders comparison	Create unified data schemas

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

- [redacted]
  - [redacted]
- [redacted]
  - [redacted]
- [redacted]
  - [redacted]
- [redacted]
  - [redacted]
- [redacted]
  - [redacted]

[redacted]

[redacted]

- [redacted]
  - [redacted]
- [redacted]
  - [redacted]
- [redacted]
  - [redacted]
  - [redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

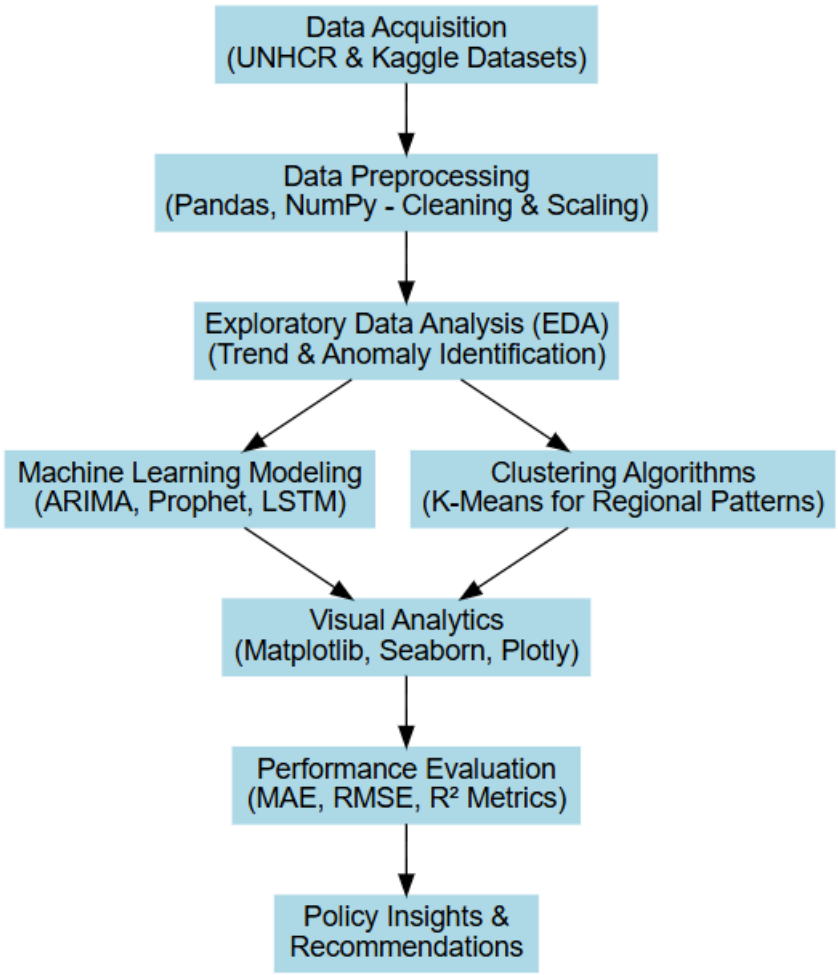
[redacted]

[redacted]

[redacted]

[redacted]

[redacted]


Data Acquisition
(UNHCR & Kaggle Datasets)
Data Preprocessing
(Pandas, NumPy - Cleaning & Scaling)
Exploratory Data Analysis (EDA)
(Trend & Anomaly Identification)
Machine Learning Modeling
(ARIMA, Prophet, LSTM)
Clustering Algorithms
(K-Means for Regional Patterns)
Visual Analytics
(Matplotlib, Seaborn, Plotly)
Performance Evaluation
(MAE, RMSE, R² Metrics)
Policy Insights &
Recommendations

[redacted]

[redacted]

```
Dataset URL: https://www.kaggle.com/datasets/unitednations/refugee-data
License(s): CC-BY-SA-4.0
Downloading refugee-data.zip to refugee_data_dataset
  0% 0.00/5.79M [00:00<?, ?B/s]
100% 5.79M/5.79M [00:00<00:00, 737MB/s]
Archive:  refugee_data_dataset/refugee-data.zip
  inflating: refugee_data_dataset/asylum_seekers.csv
  inflating: refugee_data_dataset/asylum_seekers_monthly.csv
  inflating: refugee_data_dataset/demographics.csv
  inflating: refugee_data_dataset/persons_of_concern.csv
  inflating: refugee_data_dataset/resettlement.csv
  inflating: refugee_data_dataset/time_series.csv
['persons_of_concern.csv',
 'resettlement.csv',
 'asylum_seekers.csv',
 'asylum_seekers_monthly.csv',
 'time_series.csv',
 'demographics.csv',
 'refugee-data.zip']
```

[redacted]

[redacted]

```
/tmp/ipython-input-2717402865.py:2: DtypeWarning: Columns (4) have mixed types. Specify dtype option on import or set low_memory=False.
  df = pd.read_csv("refugee_data_dataset/time_series.csv")
```

	Year	Country / territory of asylum/residence	Origin	Population type	Value
0	1951	Australia	Various/Unknown	Refugees (incl. refugee-like situations)	180000
1	1951	Austria	Various/Unknown	Refugees (incl. refugee-like situations)	282000
2	1951	Belgium	Various/Unknown	Refugees (incl. refugee-like situations)	55000
3	1951	Canada	Various/Unknown	Refugees (incl. refugee-like situations)	168511
4	1951	Switzerland	Various/Unknown	Refugees (incl. refugee-like situations)	10000
5	1951	Germany	Various/Unknown	Refugees (incl. refugee-like situations)	265000
6	1951	Denmark	Various/Unknown	Refugees (incl. refugee-like situations)	2000
7	1951	Spain	Various/Unknown	Refugees (incl. refugee-like situations)	2000
8	1951	France	Various/Unknown	Refugees (incl. refugee-like situations)	290000
9	1951	United Kingdom	Various/Unknown	Refugees (incl. refugee-like situations)	208000

[redacted]

[redacted]

[redacted]

[redacted]

	0
Year	int64
Country / territory of asylum/residence	object
Origin	object
Population type	object
Value	float64

dtype: object

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

	0
Year	0
Country / territory of asylum/residence	0
Origin	0
Population type	0
Value	4208

dtype: int64

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

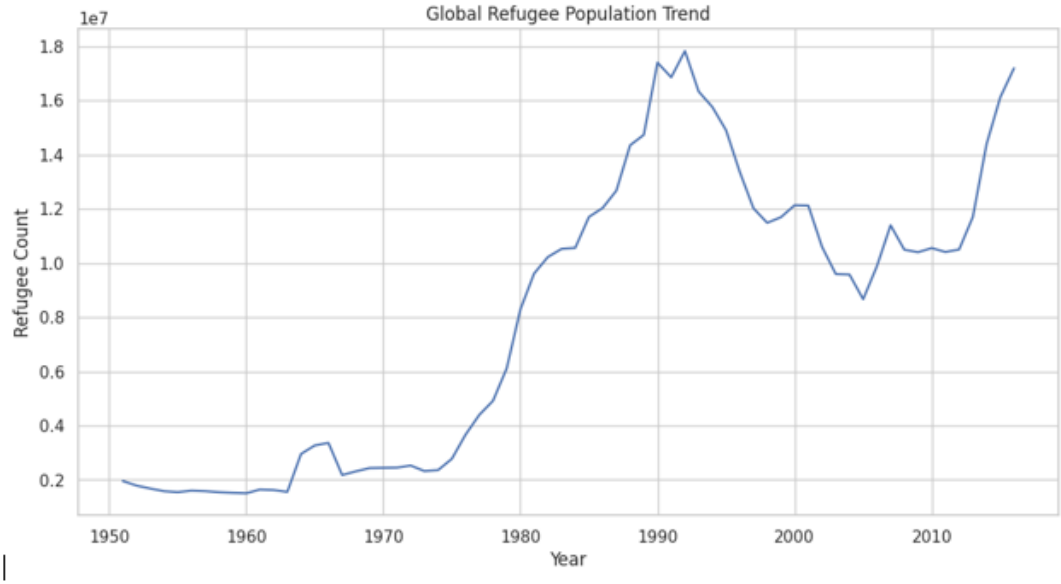
[redacted]

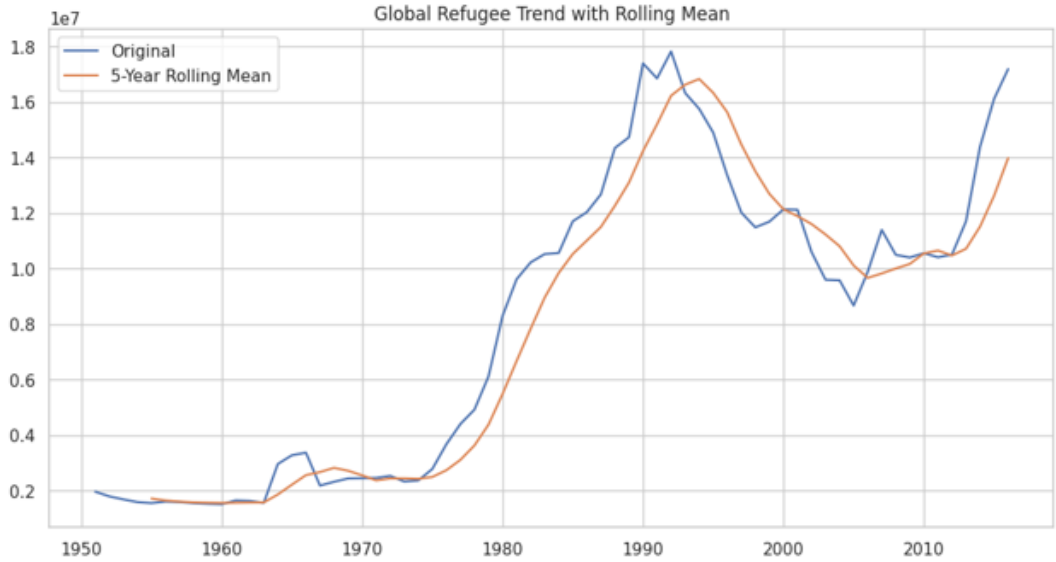
	Year	Value
0	1951	1957211.0
1	1952	1787928.0
2	1953	1679304.0
3	1954	1581628.0
4	1955	1547066.0
5	1956	1600875.0
6	1957	1582814.0
7	1958	1545310.0
8	1959	1520185.0
9	1960	1504664.0

[redacted]

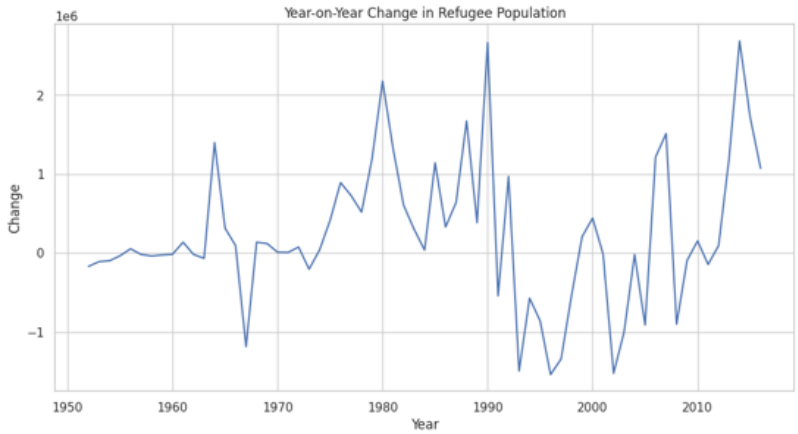
[redacted]

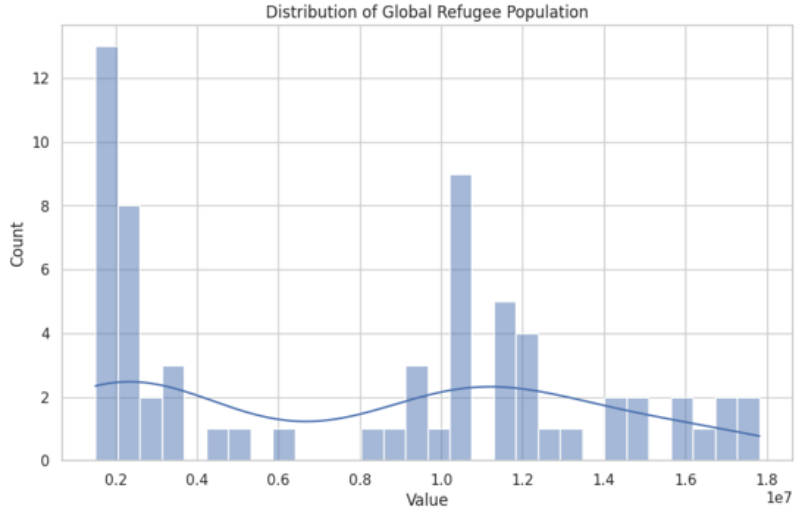
[redacted]



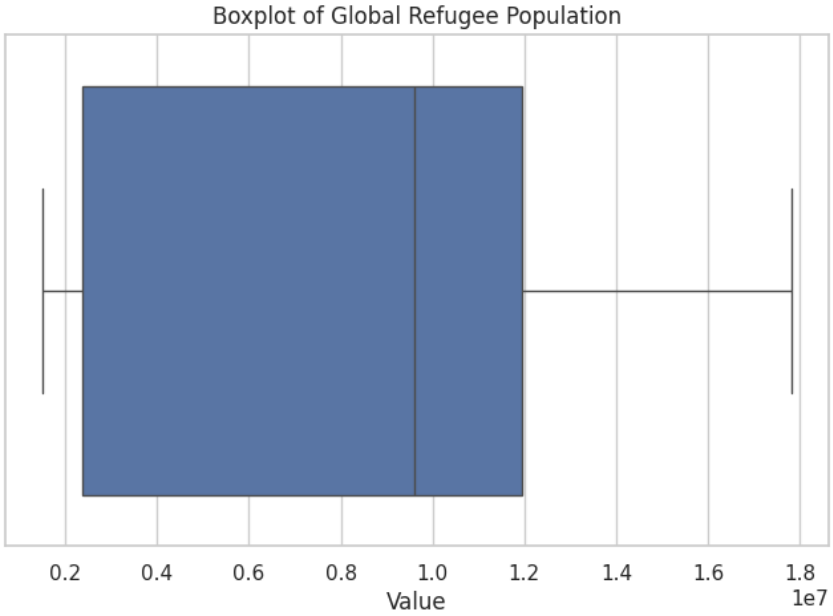


[redacted]

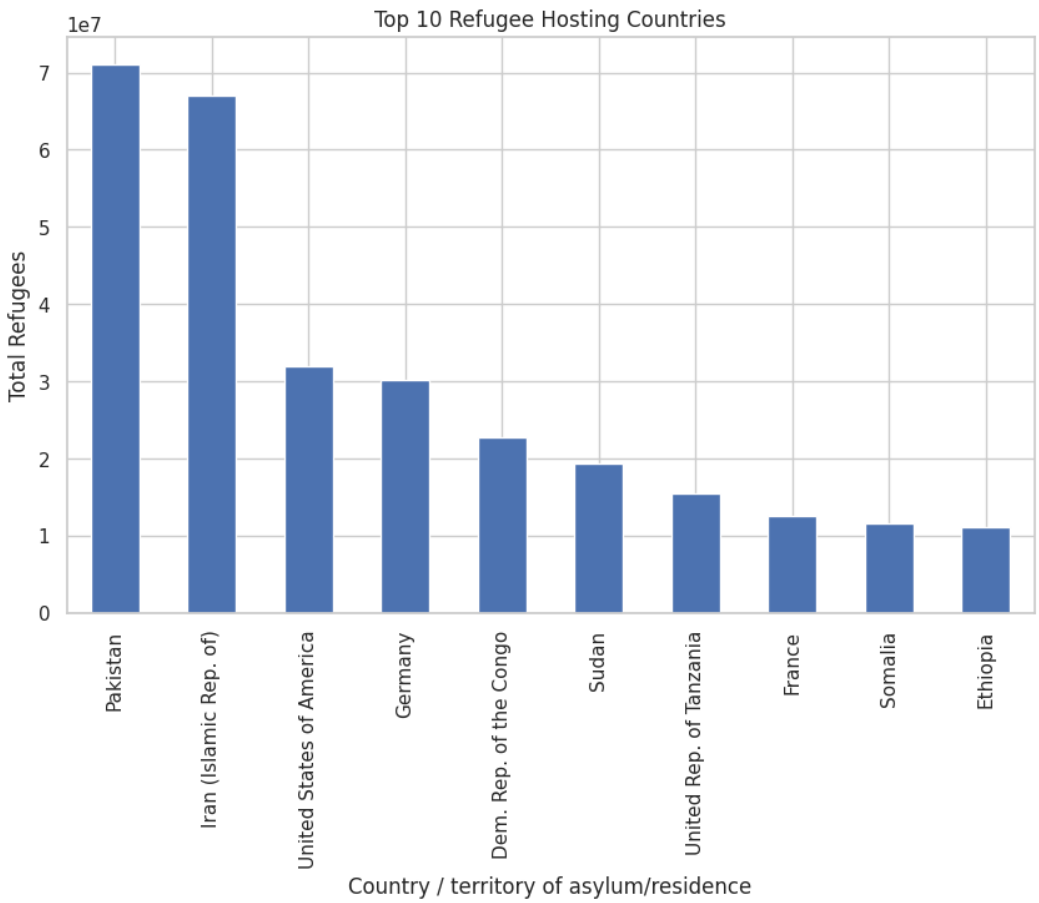




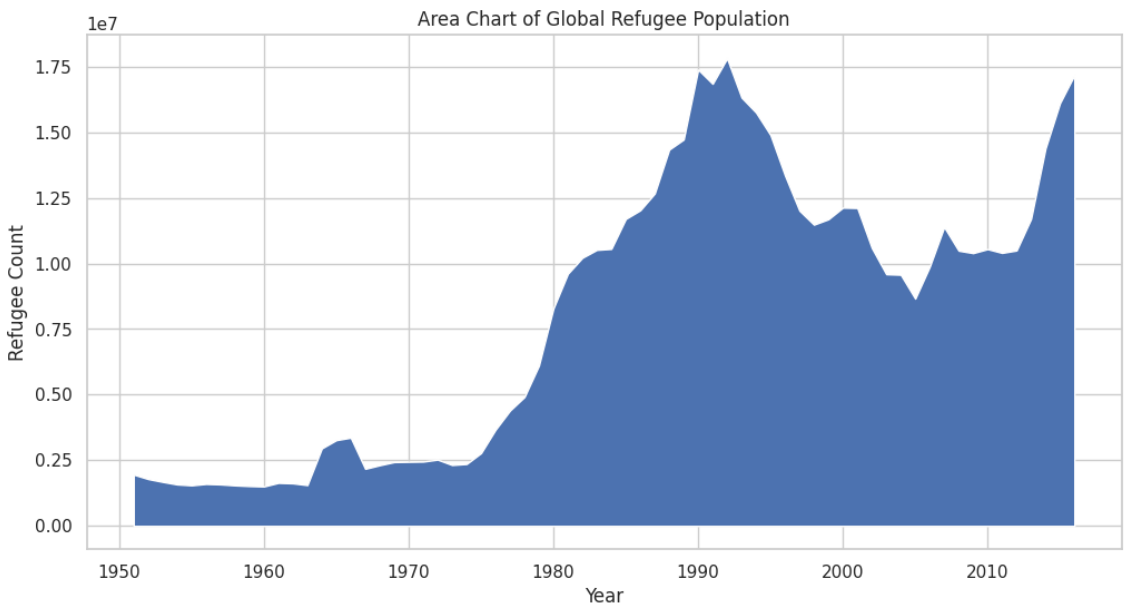
[redacted]



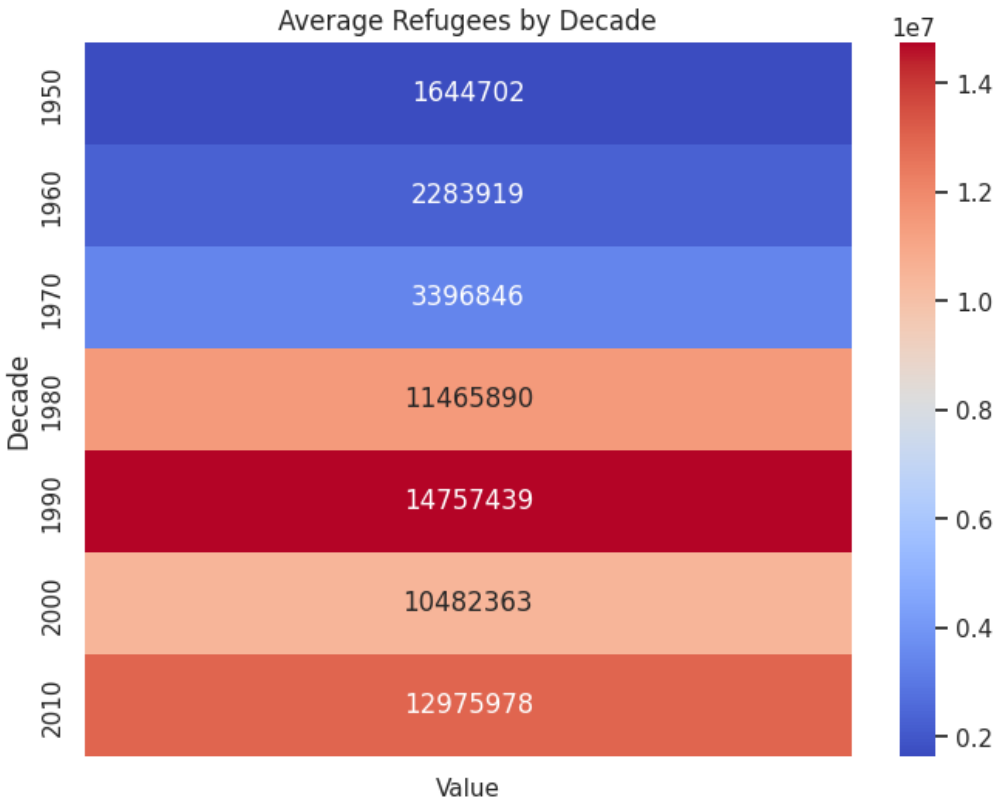
[redacted]



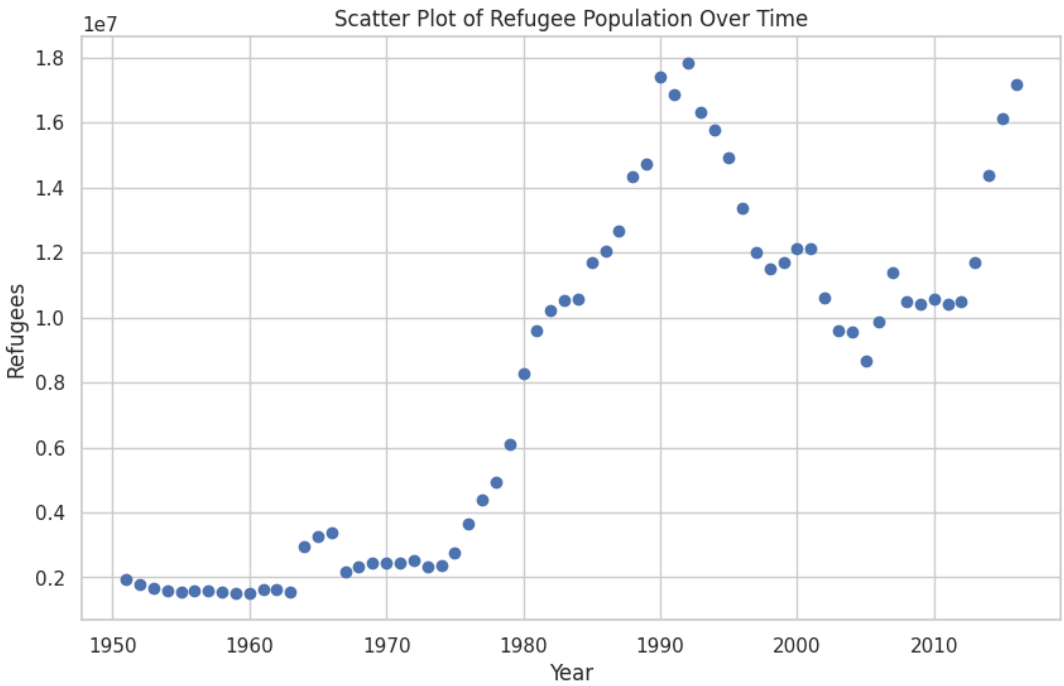
[redacted]



[redacted]



[redacted]



[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

```python
# Core libraries
import numpy as np
import pandas as pd

# Visualization
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

# Time-series models
from statsmodels.tsa.arima.model import ARIMA
from sklearn.metrics import mean_absolute_error, mean_squared_error

# Deep learning
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense
from sklearn.preprocessing import MinMaxScaler

# Settings
plt.rcParams["figure.figsize"] = (12,6)
sns.set(style="whitegrid")

print("Environment ready")
```

```python
# Reload dataset with safe memory handling
df = pd.read_csv(
    "refugee_data_dataset/time_series.csv",
    low_memory=False
)

# Convert Year to integer
df["Year"] = df["Year"].astype(int)

# Convert Value to numeric
df["Value"] = pd.to_numeric(df["Value"], errors="coerce")

# Check data types
df.dtypes
```

```python
# Remove rows where Value is missing
df = df.dropna(subset=["Value"])

# Verify missing values again
df.isnull().sum()
```

```python
# Load main time-series dataset
df = pd.read_csv("refugee_data_dataset/time_series.csv")

# Show top 10 rows
df.head(10)
```

```python
# Filter only refugee population
df_refugees = df[df["Population type"] == "Refugees (incl. refugee-like situations)"]

# Aggregate total refugees per year (global)
global_refugees = (
    df_refugees
    .groupby("Year")["Value"]
    .sum()
    .reset_index()
)

# Sort by year
global_refugees = global_refugees.sort_values("Year")

# Display top 10 rows
global_refugees.head(10)
```

```python
#  ARIMA MODEL

# Train ARIMA
arima_model = ARIMA(train["Refugee_Count"], order=(1,1,1))
arima_result = arima_model.fit()

# Forecast
arima_forecast = arima_result.forecast(steps=len(test))
arima_forecast.index = test.index

# Evaluation
arima_mae = mean_absolute_error(test["Refugee_Count"], arima_forecast)
arima_rmse = np.sqrt(mean_squared_error(test["Refugee_Count"], arima_forecast))

print("ARIMA MAE:", arima_mae)
print("ARIMA RMSE:", arima_rmse)

#  VISUALIZATION
plt.figure(figsize=(12,6))
plt.plot(train.index, train["Refugee_Count"], label="Training Data")
plt.plot(test.index, test["Refugee_Count"], label="Actual Test Data")
plt.plot(arima_forecast.index, arima_forecast, label="ARIMA Forecast")
plt.title("ARIMA Forecast vs Actual Refugee Counts")
plt.xlabel("Year")
plt.ylabel("Number of Refugees")
plt.legend()
plt.show()
```

```python
#  PROPHET MODEL

# Install prophet if not installed
!pip install -q prophet

from prophet import Prophet

# Prepare data for Prophet
prophet_df = ts_data.reset_index()
prophet_df.columns = ["ds", "y"]

# Train-test split
prophet_train = prophet_df.iloc[:train_size]
prophet_test = prophet_df.iloc[train_size:]

# Train Prophet model
prophet_model = Prophet(
    yearly_seasonality=True,
    daily_seasonality=False,
    weekly_seasonality=False
)
prophet_model.fit(prophet_train)

# Create future dataframe
future = prophet_model.make_future_dataframe(periods=len(prophet_test), freq="Y")

# Forecast
prophet_forecast = prophet_model.predict(future)
```

```python
#  LSTM MODEL

# Scale data
scaler = MinMaxScaler(feature_range=(0,1))
scaled_data = scaler.fit_transform(ts_data)

# Create sequences
def create_sequences(data, seq_length=3):
    X, y = [], []
    for i in range(len(data) - seq_length):
        X.append(data[i:i+seq_length])
        y.append(data[i+seq_length])
    return np.array(X), np.array(y)

X, y = create_sequences(scaled_data, seq_length=3)

# Train-test split
X_train, X_test = X[:train_size-3], X[train_size-3:]
y_train, y_test = y[:train_size-3], y[train_size-3:]

# Build LSTM model
model = Sequential()
model.add(LSTM(50, activation="tanh", input_shape=(X_train.shape[1], 1)))
model.add(Dense(1))
model.compile(optimizer="adam", loss="mse")

# Train model
model.fit(X_train, y_train, epochs=50, batch_size=8, verbose=0)
```

[redacted]

```python
#  COMPARISON TABLE
comparison_df = pd.DataFrame({
    "Model": ["ARIMA", "Prophet", "LSTM"],
    "MAE": [arima_mae, prophet_mae, lstm_mae],
    "RMSE": [arima_rmse, prophet_rmse, lstm_rmse]
})

display(comparison_df)

#  COMPARISON PLOT
plt.figure(figsize=(10,6))
x = np.arange(len(comparison_df["Model"]))

plt.bar(x - 0.2, comparison_df["MAE"], width=0.4, label="MAE")
plt.bar(x + 0.2, comparison_df["RMSE"], width=0.4, label="RMSE")

plt.xticks(x, comparison_df["Model"])
plt.xlabel("Forecasting Model")
plt.ylabel("Error Value")
plt.title("Model Performance Comparison (MAE & RMSE)")
plt.legend()
plt.show()
```

[redacted]

```python
#  DASHBOARD LOGIC
def render_dashboard(*args):
    with out:
        clear_output(wait=True)

        country = country_dd.value
        model_name = model_rb.value

        data = df[df["Country / territory of asylum/residence"] == country]
        ts = data.groupby("Year")["Value"].sum().reset_index()

        if len(ts) < 10:
            print("Not enough data for forecasting.")
            return

        train_size = int(len(ts) * 0.8)
        train, test = ts.iloc[:train_size], ts.iloc[train_size:]

        #  FORECAST
        if model_name == "ARIMA":
            forecast = ARIMA(train["Value"], order=(1,1,1)).fit().forecast(len(test))

        elif model_name == "Prophet":
            p = ts.rename(columns={"Year":"ds","Value":"y"})
            p["ds"] = pd.to_datetime(p["ds"], format="%Y")
            m = Prophet()
            m.fit(p.iloc[:train_size])
            future = m.make_future_dataframe(periods=len(test), freq="YE")
            forecast = m.predict(future)["yhat"].iloc[-len(test):]
```

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

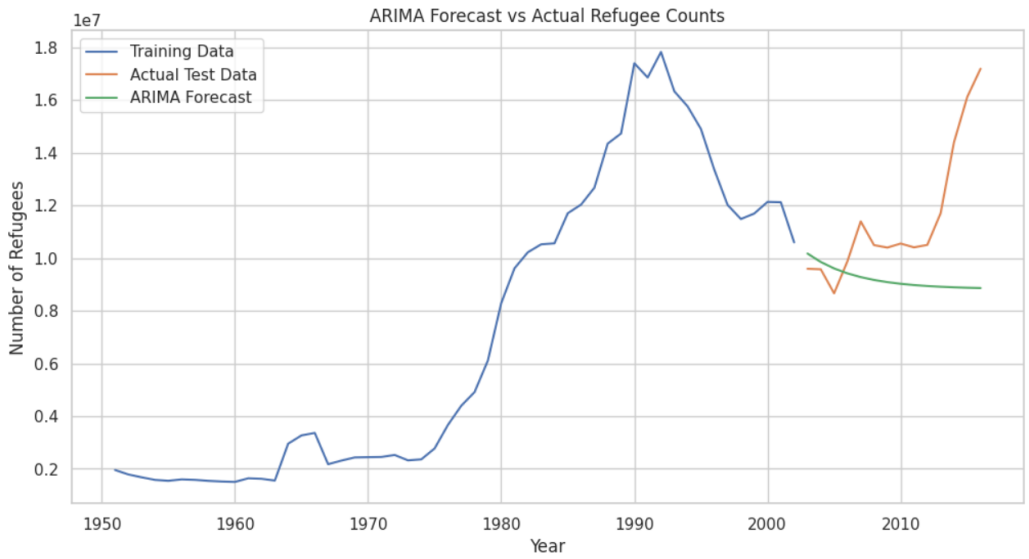
[redacted]

[redacted]

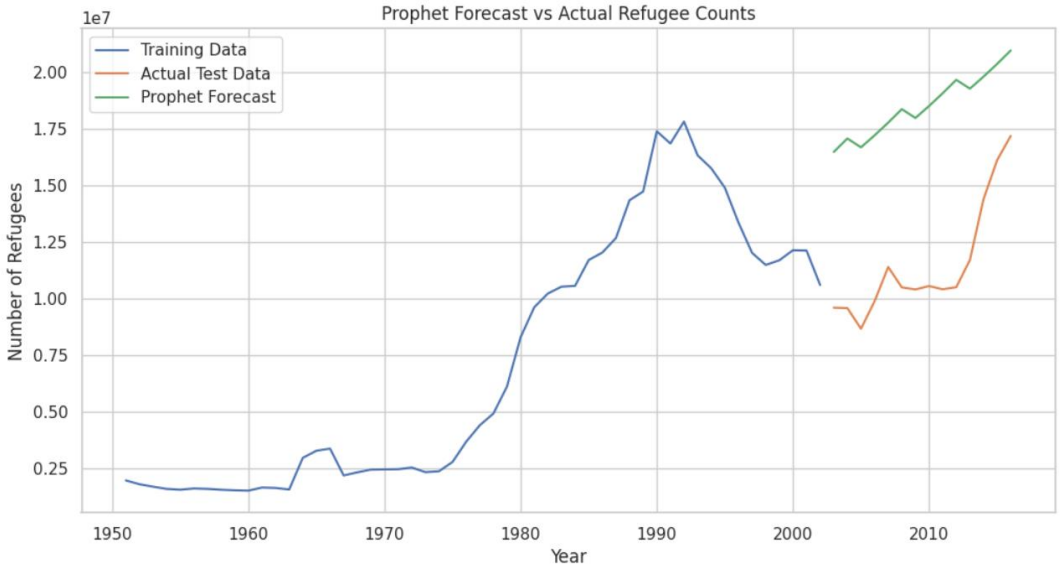
[redacted]

[redacted]

[redacted]


ARIMA Forecast vs Actual Refugee Counts
1e7
Training Data
Actual Test Data
ARIMA Forecast
Number of Refugees
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
1950
1960
1970
1980
1990
2000
2010
Year

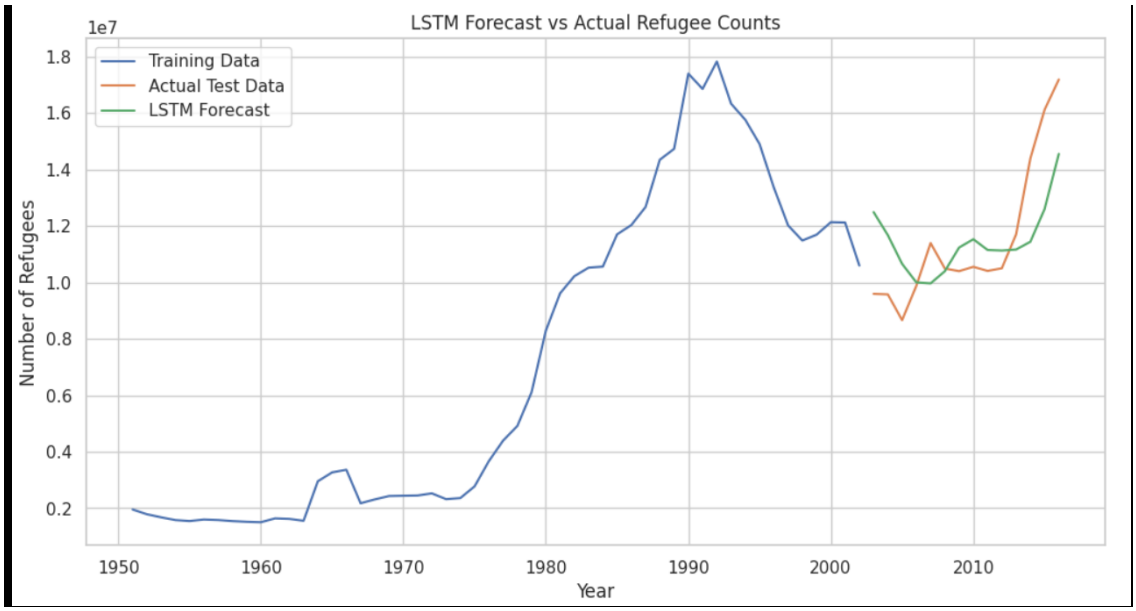
[redacted]



[redacted]

[redacted]

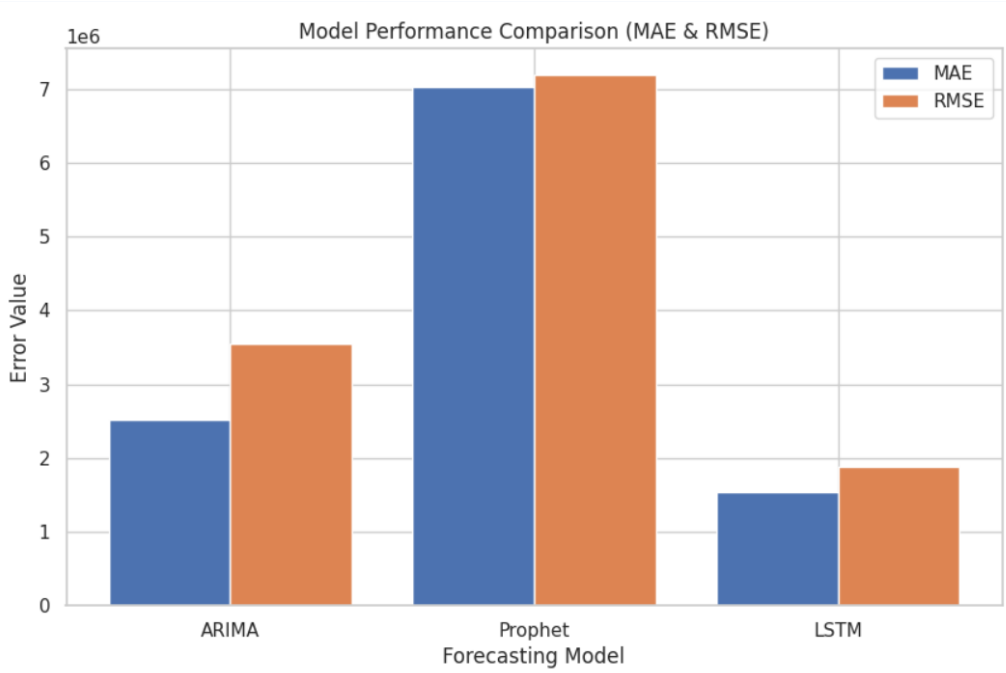
[redacted]



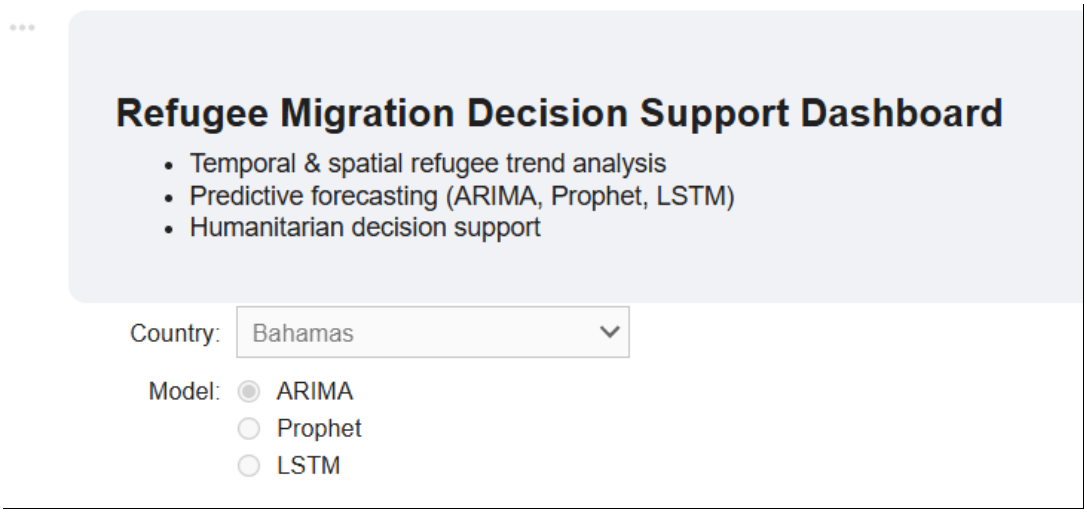
[redacted]

[redacted]

[redacted]



[redacted]


Refugee Migration Decision Support Dashboard
Temporal & spatial refugee trend analysis
Predictive forecasting (ARIMA, Prophet, LSTM)
Humanitarian decision support
Country:
Bahamas
Model:
ARIMA
Prophet
LSTM

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]